

Energy-Energy Correlators at Strong Coupling

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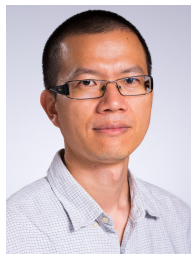
Based on 2601.05312 with Bo Wang and Congkao Wen
2607.21139 with Max Jackson, Bo Wang, and Congkao Wen



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Why energy correlators

- Measured in experiments: e^+e^- colliders, ep colliders, pp colliders.
- Infrared safe; bridge formal theory and experiments.

[Moult, Zhu, 2025 review]

Why $\mathcal{N} = 4$ super Yang–Mills

- AdS/CFT allows analytic computation at strong coupling.

Question

- How does an AdS string amplitude determine the angular energy correlator?

Main Takeaway

- We express EEC as a worldsheet integral.
- This circumvents the complexity of the amplitude itself and helps obtain the EEC in a more efficient way.
- The EEC transformation and the strong coupling expansion *might* not commute.

[Wang, Wen, LR; Jackson, Wang, Wen, LR]

Energy flux operator

$$\mathcal{E}(\hat{n})|X\rangle = \sum_{k \in X} k^0 \delta^2(\Omega_n - \Omega_k) |X\rangle.$$

[Sterman]

An operator definition

$$\mathcal{E}(\hat{n}) = \frac{1}{4} \int_{-\infty}^{\infty} du \lim_{r \rightarrow \infty} r^2 T_{\mu\nu}(u+r, r\hat{n}) \bar{n}^\mu \bar{n}^\nu, \quad \bar{n} = (1, -\hat{n}).$$

where $n = (1, \vec{n})$ is null.

[Sveshnikov, Tkachov; Korchemsky, Sterman; Hofman, Maldacena]

[Kologlu, Kravchuk, Simmons-Duffin, Zhiboedov]

The energy-energy correlator

$$\text{EEC}(\xi) = \frac{8\pi^2}{q^2 \sigma_{\text{tot}}(q)} \int d^4x e^{iq \cdot x} \langle \mathcal{O}(x) \mathcal{E}(n_1) \mathcal{E}(n_2) \mathcal{O}(0) \rangle.$$

$$\sigma_{\text{tot}}(q) = \int d^4x e^{iq \cdot x} \langle \mathcal{O}(x) \mathcal{O}(0) \rangle.$$

- A local source creates the state with total momentum q .
- Two detectors measure the angular correlation of energy flux.
- In the rest frame,

$$\xi = \frac{1 - \cos \theta}{2} = \sin^2 \frac{\theta}{2}, \quad 0 \leq \xi \leq 1.$$

[Hofman, Maldacena; Belitsky, Hohenegger, Korchemsky, Sokatchev, Zhiboedov]

Energy and momentum conservation

$$\int_0^1 d\xi \text{EEC}(\xi) = 1, \quad \int_0^1 d\xi (1 - 2\xi) \text{EEC}(\xi) = 0.$$

- The angular dependence is naturally expanded in Legendre multipoles.

$$\text{EEC}(\xi) = \sum_{s=0}^{\infty} c_s P_s(1 - 2\xi), \quad c_0 = 1, \quad c_1 = 0.$$

- The first unconstrained angular moment is c_2 .

[Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov]

- We choose the source to be a half-BPS scalar operator $\mathcal{O}_p$.
- For $p = 2$, $\mathcal{O}_2$ belongs to the stress-tensor multiplet. The two $\mathcal{O}_2$ insertions are related to energy detectors by supersymmetry and light-ray transforms.
- The local input is the four-point correlator

$$\langle \mathcal{O}_p \mathcal{O}_2 \mathcal{O}_2 \mathcal{O}_p \rangle.$$

[Belitsky, Hohenegger, Korchemsky, Sokatchev, Zhiboedov; Jackson, Wang, Wen, LR]

Examples of EEC

- Weak coupling:

$$\text{EEC}_2(\xi) = \tfrac{1}{2}[\delta(\xi) + \delta(1 - \xi)] + \mathcal{O}(\lambda).$$

- Strong coupling:

$$\text{EEC}(\xi) = 1 + \mathcal{O}(\lambda^{-1}).$$

[Hofman, Maldacena]

- Weak coupling: fruitful knowledge at high multiplicity and high loop order.

[Dixon, Luo, Shtabovenko, Yang, Zhu; Henn, Sokatchev, Yan, Zhiboedov;
He, Jiang, Yang, Zhang; Volovich, Wu, Yan]

- Numerical bootstrap: non-perturbative bounds and finite coupling information from CFT data.

[Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov]

- Strong coupling: the four-point AdS amplitude is known up to first curvature for general p and second curvature correction for $p = 2$.

[Alday, Hansen, Silva; Alday, Hansen; Fardelli, Hansen, Silva]

After analytic continuation and the light-ray transform,

$$\text{EEC}_p(\xi) = \int \frac{ds dt}{(2\pi i)^2} \underbrace{\mathcal{M}_p(s, t)}_{\substack{\text{dynamics} \\ \text{(amplitude)}}} \underbrace{\mathcal{K}_p(t; \xi)}_{\substack{\text{detector/angular} \\ \text{dependence}}} .$$

[Belitsky, Hohenegger, Korchemsky, Sokatchev, Zhiboedov]

- $\mathcal{M}_p(s, t)$ is the Mellin amplitude of the four-point correlator.
- The detector kernel contains the angular dependence. For $p = 2$,

$$\mathcal{K}_2(t; \xi) = \frac{-\pi t^2 (t-2)^2}{128 \xi^3 \sin(\pi t/2)} \left(\frac{\xi}{1-\xi} \right)^{1+t/2} .$$

- The kernel is independent of s . The s -integral is the essential dynamical step.

Note that

$$\text{EEC}_p(\xi) = \int_{i\mathbb{R}} \frac{ds dt}{(2\pi i)^2} \underbrace{\mathcal{M}_p(s, t)}_{\substack{\text{dynamics} \\ \text{(amplitude)}}} \underbrace{\mathcal{K}_p(t; \xi)}_{\substack{\text{detector/angular} \\ \text{dependence}}}.$$

Define

$$Q_{n,p}(\xi) = \int \frac{dt}{2\pi i} t^n \mathcal{K}_p(t; \xi).$$

with Q being a linear combination of Legendre polynomials:

$$Q_{0,2} = P_2(1 - 2\xi), \quad Q_{1,2} = P_2(1 - 2\xi) + 3P_3(1 - 2\xi), \dots$$

Then we have the expansion

$$\text{EEC}_p(\xi) = 1 + \sum_{k=0}^{\infty} \lambda^{-1-k/2} \sum_{n=0}^k c_{k,n}(p) Q_{k-n,p}(\xi).$$

For $p = 2$, the low-energy expansion begins with

$$\mathcal{M}_2(s, t) = \underbrace{\frac{8}{(s-2)(t-2)(\tilde{u}-2)}}_{\text{supergravity}} + \underbrace{\frac{120\zeta(3)}{\lambda^{3/2}} + \dots}_{\text{stringy corrections}}, \quad s + t + \tilde{u} = 4.$$

[Rastelli, Zhou; Alday, Hansen, Silva]

- SUGRA gives $\text{EEC}(\xi) = 1$.
- The stringy terms in the low-energy expansion are polynomials in s, t .
- Their s -integrals are not well-defined term by term on an infinite vertical contour.

We must reorganize the summation and integration.

[Gonçalves]

Relate the AdS amplitude to the Mellin amplitude through a Borel representation:

[Penedones]

$$\mathcal{M}_p(s, t) = \frac{1}{2\lambda^{3/2}\Gamma(p-1)\Gamma(p)} \int_0^\infty d\beta \beta^{p+3} e^{-\beta} A(\beta\hat{s}, \beta\hat{t}),$$
$$\hat{s} = \frac{s - 2p/3}{2\sqrt{\lambda}}, \quad \hat{t} = \frac{t - 2p/3}{2\sqrt{\lambda}}, \quad S = \beta\hat{s}, \quad T = \beta\hat{t}.$$

- In Mellin space, fixed orders in the strong-coupling expansion are polynomial low-energy data.
- In AdS amplitude space, each fixed-curvature object keeps the resummed string-energy dependence:

$$A^{(k)}(S, T) \lesssim \frac{1}{|S|^2}, \quad S \rightarrow i\infty$$

[Maldacena, Shenker, Stanford; Costa, Goncalves, Penedones; Alday, Hansen]

AdS curvature expansion

$$A(S, T) = \sum_{k=0}^{\infty} \lambda^{-k/2} A^{(k)}(S, T), \quad \frac{1}{\sqrt{\lambda}} = \frac{\alpha'}{R^2}, \quad S + T + U = 0.$$

[Alday, Hansen, Silva; Alday, Hansen]

The leading term is the full flat-space Virasoro-Shapiro amplitude:

$$A^{(0)}(S, T) = -\frac{\Gamma(-S)\Gamma(-T)\Gamma(S+T)}{\Gamma(1+S)\Gamma(1+T)\Gamma(1-S-T)}.$$

[Virasoro; Shapiro]

The essential object is

$$I^{(k)}(T) = \int \frac{dS}{2\pi i} A^{(k)}(S, T).$$

Coefficients of $I^{(k)}(T)$ feed the data of EEC expansion in Q .

From $I(T)$ to EEC expansion data

$$I^{(r)}(T) = \sum_{i \geq 0} e_i^{(r)} T^i.$$

$$[Q_{j,p}]_{\lambda^{-k/2-1}} = \frac{1}{\Gamma(p-1)\Gamma(p)} \sum_{i=j}^k \frac{\Gamma(p+3+i)}{2^i} \binom{i}{j} \left(-\frac{2p}{3}\right)^{i-j} e_i^{(k-i)}.$$

	$Q_{0,p}$	$Q_{1,p}$	$Q_{2,p}$
λ^{-1}	$\{e_0^{(0)}\}$	$\{\}$	$\{\}$
$\lambda^{-3/2}$	$\{e_0^{(1)}, e_1^{(0)}\}$	$\{e_1^{(0)}\}$	$\{\}$
λ^{-2}	$\{e_0^{(2)}, e_1^{(1)}, e_2^{(0)}\}$	$\{e_1^{(1)}, e_2^{(0)}\}$	$\{e_2^{(0)}\}$

Closing the S contour

The basic object is a vertical contour integral,

$$I^{(k)}(T) = \int_{\varepsilon - i\infty}^{\varepsilon + i\infty} \frac{dS}{2\pi i} A^{(k)}(S, T).$$

- With the Regge falloff, the contour can be closed in the appropriate half-plane.
- The EEC data can then be read from a sum over pole residues.
- This turns the problem into summing over the massive-pole data of the AdS amplitudes.

$$\text{EEC}^{(k)} \sim \sum_{\text{massive poles}} \text{Res } A^{(k)}.$$

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The leading strong-coupling moments are finite:

$$c_3 = \frac{360\zeta(3)}{\lambda^{3/2}} + \cdots, \quad c_4 = \frac{684\pi^4}{7\lambda^2} + \cdots.$$

- The strong-coupling correction is reconstructed from stringy operators.
- The problem appears when the large- λ expansion is taken before resumming the higher-order data.
- The higher order of the residue summations diverge!

[Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov]

Higher orders of the residue sum**Example: the T^1 coefficient of $I^{(1)}(T)$**

$$\text{raw residue sum} = -2\zeta(2)H_\Delta + C_1 + o(1), \quad \Delta \rightarrow \infty,$$

$$C_1 = 2\zeta(2) - 4\zeta(3) + \frac{53}{2}\zeta(4), \quad H_\Delta = \sum_{\delta=1}^{\Delta} \frac{1}{\delta}.$$

The raw residue sum diverges!

- So the formal step “close contour then sum residues” cannot be used naively after the large- λ expansion.

Keep the imaginary-axis integral

$$I^{(k)}(T) = \int_{\varepsilon-i\infty}^{\varepsilon+i\infty} \frac{dS}{2\pi i} A^{(k)}(S, T)$$

- At fixed curvature order, $A^{(k)} \sim 1/|S|^2$ on the imaginary S axis.
- The strong-coupling expansion of the AdS amplitude is well-defined.
- Therefore the S integral itself should give a finite answer for the EEC data.

[large λ expansion, residue sum] $\neq 0$

[large λ expansion, residue sum] $\neq 0$

Instead of the divergence, the non-commutativity also affect the finite terms:

Example: the T^0 coefficient of $I^{(1)}(T)$

The OPE expansion includes:

$$A^{(1)}(S, 0) = (\text{safe terms}) - \sum_{\delta=1}^{\infty} Z_2(\delta-1) \left[\frac{1}{(S-\delta)^2} + \frac{1}{(S+\delta)^2} \right].$$

$$\sum_{\delta} \int \frac{dS}{2\pi i} (\cdots) = 0, \quad \int \frac{dS}{2\pi i} \sum_{\delta} (\cdots) = -\zeta(2).$$

The direct imaginary-axis integral is finite, but reconstructing it from all massive-pole data is cumbersome.

At fixed curvature order, we use two simpler representations:

- ① Worldsheet integrals over a disk and its boundary.
- ② Resummation of the low-energy Wilson coefficients.

[Wang, Wen, LR; Jackson, Wang, Wen, LR]

- Both match with integral along the imaginary axis.
- A guaranteed finite answer.

Worldsheet formulation for AdS Virasoro-Shapiro amplitudes:

$$A^{(k)}(S, T) = \int d^2z |z|^{-2S-2} |1-z|^{-2T-2} G_{\text{tot}}^{(k)}(z, \bar{z}; S, T).$$

[Alday, Hansen, Silva; Alday, Hansen; Fardelli, Hansen, Silva]

- $G_{\text{tot}}^{(k)}$ contains single-valued multiple polylogarithms of weight $3k$.
- First curvature: weight three, including general Kaluza-Klein external states.
- Second curvature: weight six, known for $p = 2$.

Compute $I^{(k)}(T)$ directly by performing the S -integral inside the worldsheet representation.

The S -integral in $I^{(k)}(T)$ acts on terms of the form

$$J_j(T) = \int_{\varepsilon - i\infty}^{\varepsilon + i\infty} \frac{dS}{2\pi i} \int d^2z S^j |z|^{-2S-2} |1-z|^{-2T-2} J(z, \bar{z}; T),$$

with $\varepsilon > 0$, the S -integral localizes the worldsheet integral:

$$j < 0 : \quad \text{contour deformation} \quad \implies \quad |z| < 1,$$

$$j \geq 0 : \quad \delta^{(j)}(\log |z|^2) \quad \implies \quad |z| = 1.$$

- ① Expand the moduli-space integrand at the desired curvature order.
- ② Perform the S -integral first.
- ③ Evaluate disk and boundary-circle integrals and expand over T .

[Wang, Wen, LR; Jackson, Wang, Wen, LR]

Virasoro-Shapiro example revisited

$$\begin{aligned} I^{(0)}(T) &= \int_{|z|<1} d^2z |1-z|^{-2T-2} (-\log |z|^2) \\ &= \frac{1}{T^2} \left[\frac{\Gamma(1-2T)}{\Gamma(1-T)^2} - 1 \right]. \end{aligned}$$

$$I^{(0)}(T) = \zeta(2) + 2\zeta(3)T + \frac{19}{4}\zeta(4)T^2 + \cdots.$$

$$c_{0,0} = 24\zeta(2), \quad c_{1,0} = 120\zeta(3), \quad c_{2,0} = 855\zeta(4).$$

[Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov; Wang, Wen, LR]

Examples from the first curvature correction

These are the two kinds of worldsheet terms needed for $I^{(1)}(T)$.

- Disk integral

$$\begin{aligned} I_1^{(1)}(T) &= \frac{p^2 T}{36} \int_{|z| < 1} d^2 z |z|^{2T-2} |1 - z|^{-2T-2} \log^3 \frac{|z|^2}{|1 - z|^2} \\ &= -\frac{p^2 T}{36} \frac{\partial^3}{\partial T^3} \frac{\Gamma(-2T)}{\Gamma(1 - T)^2}. \end{aligned}$$

- Circle integral, $z = e^{i\theta}$

$$I_3^{(1)}(T) = \oint_{|z|=1} \frac{d\theta}{2\pi} |1 - z|^{-2T-2} (2\zeta(3) - \text{Li}_3(z) - \text{Li}_3(\bar{z})).$$

[Alday, Hansen; Fardelli, Hansen, Silva; Jackson, Wang, Wen, LR]

First curvature correction

- Disk integrals: logarithms and derivatives of Gamma-function integrals.
- Boundary-circle integrals: weight-three polylogarithms.

[Alday, Hansen, Silva; Fardelli, Hansen, Silva]

For $p = 2$, the regular expansion is

$$\begin{aligned} I^{(1)}(T) &= \zeta(2) - \frac{10}{3}\zeta(3) \\ &\quad + \left[-2\zeta(2) + 4\zeta(3) - \frac{53}{2}\zeta(4) \right] T + \mathcal{O}(T^2). \end{aligned}$$

This gives

$$\begin{aligned} c_{1,1} &= 24\zeta(2) - 240\zeta(3), \\ c_{2,1} &= -120\zeta(2) + 240\zeta(3) - 3870\zeta(4). \end{aligned}$$

[Wang, Wen, LR; Jackson, Wang, Wen, LR]

Second curvature correction, $p = 2$

- The worldsheet correlator contains weight-six single-valued multiple polylogarithms.
- To determine the EEC through λ^{-2} , only $I^{(2)}(0)$ is needed.
- This constant receives a contribution only from the unit-circle integral. The regular disk contribution starts at order T .

[Alday, Hansen; Jackson, Wang, Wen, LR]

$$I^{(2)}(0) = 6\zeta(2) - \frac{13}{2}\zeta(3) + \frac{263}{6}\zeta(4) \\ + \zeta(2)\zeta(3) - \frac{5}{3}\zeta(5).$$

EECs from Wilson coefficients

$$\begin{aligned}
\mathcal{M}_2(s, t) &= \mathcal{M}_{\text{SUGRA}}(s, t) \\
&+ \sum_{a, b \geq 0} \frac{\Gamma(2a + 3b + 6)}{8^{a+b} \lambda^{3/2 + a + 3b/2}} (s_1^2 + s_2^2 + s_3^2)^a (s_1 s_2 s_3)^b \\
&\times [\alpha_{a, b}^{(0)} + \lambda^{-1/2} \alpha_{a, b}^{(1)} + \dots], \\
s_1 &= s - \frac{4}{3}, \quad s_2 = t - \frac{4}{3}, \quad s_1 + s_2 + s_3 = 0.
\end{aligned}$$

[Alday, Hansen, Silva; Alday, Hansen]

- Extract the required power of t .
- Resum the Wilson coefficients before performing the unbounded s -integral.

The leading example

$$c_{0,0} = \int \frac{ds_1}{2\pi i} \sum_{a=0}^{\infty} \frac{\Gamma(2a+6)}{4^a} \zeta(2a+3) s_1^{2a}.$$

[Gonçalves]

A useful modified Borel representation is

$$\zeta(n) = \frac{2^{n-1}}{\Gamma(n+1)} \int_0^{\infty} d\omega \frac{\omega^n}{\sinh^2 \omega}, \quad \text{Re } n > 1.$$

[Arutyunov, Dorigoni, Savin; Dorigoni, Green, Wen]

- Sum the infinite low-energy series using this representation.
- The contour integral gives $c_{0,0} = 24\zeta(2)$, in agreement with the worldsheet calculation.

Analytic continuation and localization

$$\begin{aligned}\mathcal{I} &= \int \frac{ds_1}{2\pi i} \sum_{a \geq 0} \frac{f(a)\Gamma(2a+p)}{4^a} \zeta(2a+q, \vec{w}) s_1^{2a+m} \\ &= 2^m f\left(-\frac{m+1}{2}\right) \Gamma(p-m-1) \zeta(q-m-1, \vec{w}).\end{aligned}$$

- The resummed integral localizes the summation variable to $a = -(m+1)/2$.

[Jackson, Wang, Wen, LR]

A term that survives the limit

The first-curvature Wilson coefficient $\alpha_{a,0}^{(1)}$ contains

$$-(2a+1)\zeta(2a+2, 2).$$

At the localized value,

$$\lim_{\varepsilon \rightarrow 0} -\varepsilon \zeta(1+\varepsilon, 2) = -\zeta(2), \quad a = -\frac{1}{2} + \frac{\varepsilon}{2}.$$

In the S -integral on summation over massive poles:

$$\int \frac{dS}{2\pi i} \sum_{\delta=1}^{\infty} Z_2(\delta-1) \left[\frac{1}{(S-\delta)^2} + \frac{1}{(S+\delta)^2} \right] = -\zeta(2)$$

The result also agrees with the worldsheet calculation.

[Alday, Hansen, Silva; Wang, Wen, LR; Jackson, Wang, Wen, LR]

General half-BPS sources

$$\begin{aligned} \text{EEC}_p(\xi) = & 1 + \frac{c_{0,0}(p)}{\lambda} Q_{0,p}(\xi) \\ & + \frac{1}{\lambda^{3/2}} [c_{1,0}(p) Q_{1,p}(\xi) + c_{1,1}(p) Q_{0,p}(\xi)] + \cdots, \end{aligned}$$

$$c_{0,0}(p) = \frac{\Gamma(p+3)}{\Gamma(p-1)\Gamma(p)} \zeta(2), \quad c_{1,0}(p) = \frac{\Gamma(p+4)}{\Gamma(p-1)\Gamma(p)} \zeta(3),$$

$$c_{1,1}(p) = \frac{\Gamma(p+3)}{\Gamma(p-1)\Gamma(p)} [\zeta(2) - (p^2 + 2p + 2)\zeta(3)].$$

[Jackson, Wang, Wen, LR]

For $p = 2$, through order λ^{-2}

$$\begin{aligned} \text{EEC}(\xi) = & 1 + \frac{24\zeta(2)}{\lambda} Q_{0,2}(\xi) \\ & + \frac{1}{\lambda^{3/2}} [120\zeta(3)Q_{1,2}(\xi) + (24\zeta(2) - 240\zeta(3))Q_{0,2}(\xi)] \\ & + \frac{1}{\lambda^2} [c_{2,0}Q_{2,2}(\xi) + c_{2,1}Q_{1,2}(\xi) + c_{2,2}Q_{0,2}(\xi)] + \cdots, \end{aligned}$$

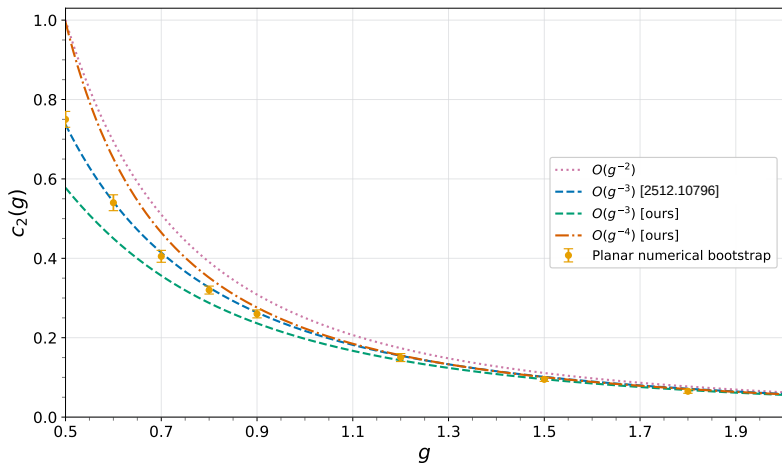
$$c_{2,0} = 855\zeta(4),$$

$$c_{2,1} = -120\zeta(2) + 240\zeta(3) - 3870\zeta(4),$$

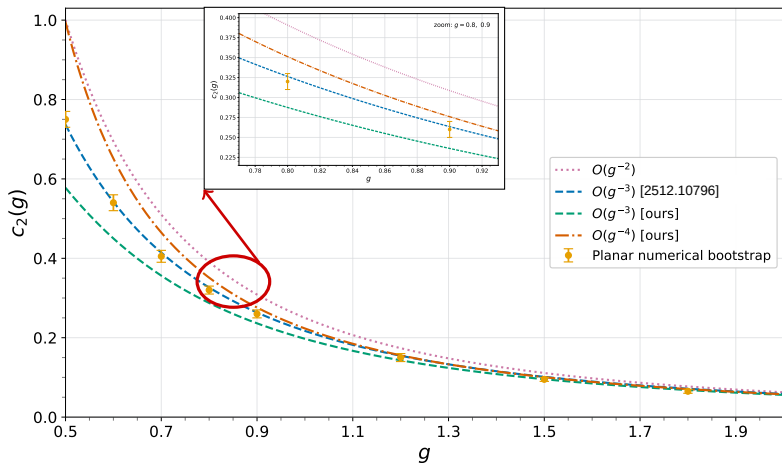
$$c_{2,2} = 304\zeta(2) - 476\zeta(3) + 24\zeta(2)\zeta(3) + 4692\zeta(4) - 40\zeta(5).$$

Worldsheet integrals and the regulated low-energy calculation agree, as well as the S -integral along the imaginary axis numerically.

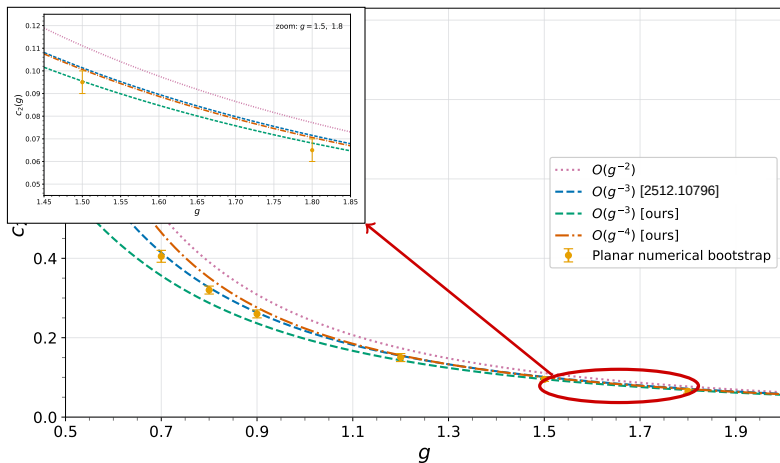
[Wang, Wen, LR; Jackson, Wang, Wen, LR]



[Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov; Jackson, Wang, Wen, LR]



[Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov; Jackson, Wang, Wen, LR]



[Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov; Jackson, Wang, Wen, LR]

Summary

- EECs in $\mathcal{N} = 4$ SYM are closely related to AdS Virasoro-Shapiro amplitudes.
- Worldsheet disk and boundary integrals provide a direct calculation.
- Resummation of Wilson coefficients gives an independent analytic cross-check.
- Both approach guarantee a finite result, therefore should work at any higher orders.

Open questions

- Can we bootstrap moduli space integrand for EEC directly without AdS amplitudes?
- Numerical bootstrap to more precise value.
- EECs with heavy states and large charge.
- Less supersymmetric theories?

Thank you!

The remaining EEC integrals

$$\text{EEC}_p(\xi) = \frac{1}{\lambda \Gamma(p-1) \Gamma(p)} \int \frac{dt}{2\pi i} \int_0^\infty d\beta e^{-\beta} \beta^{p+2} I(\beta \hat{t}) \mathcal{K}_p(t; \xi).$$

$$\hat{t} = \frac{t - 2p/3}{2\sqrt{\lambda}}, \quad \int_0^\infty d\beta e^{-\beta} \beta^{p+2+i} = \Gamma(p+3+i).$$

- Expand $I(T)$ to the required order.
- Include the shift in $\hat{t}$ before using $Q_{n,p}$.
- Treat the massless or SUGRA singular contribution separately from the regular string expansion.

[Wang, Wen, LR; Jackson, Wang, Wen, LR]

First curvature: cutoff dependence

Raw simple-pole residue sums, as $\Delta \rightarrow \infty$:

$$T^1 : -2\zeta(2)H_\Delta + C_1 + o(1), \quad T^2 : -2\zeta(2)H_\Delta^2 + C_2 + o(1).$$

$$C_1 = 2\zeta(2) - 4\zeta(3) + \frac{53}{2}\zeta(4),$$

$$C_2 = -4\zeta(2) - 6\zeta(4) + 32\zeta(2)\zeta(3) + 8\zeta(3) + 74\zeta(5).$$

After finite-axis integration and mass summation, the T^1 contributions are

$$\text{simple:} \quad 2\zeta(2)(\log Y + \gamma_E) - C_1 + o(1),$$

$$\text{double:} \quad -2\zeta(2)(\log Y + \gamma_E) + o(1).$$

- Their sum tends to $-C_1$; the raw residue sum still diverges.
- At T^2 , squared logarithms cancel between simple and double poles.
- These are ordered asymptotics, not exact finite- (Y, Δ) identities.

Second curvature: cancellation between pole orders

At $T = 0$, keep common cutoffs. The divergent coefficients are

Pole order	simple	double	triple	fourth
$\zeta(4)YH_{\Delta}/\pi$	-7	2	5	0
$\zeta(2)(\log Y + \gamma_E)$	$-29/6$	$43/6$	$-4/3$	-1

$$-7 + 2 + 5 = 0, \quad -\frac{29}{6} + \frac{43}{6} - \frac{4}{3} - 1 = 0.$$

- The first row concerns mass-cutoff asymptotics at fixed Y ; the second concerns the subsequent large- Y expansion.
- Fifth, sixth and seventh pole orders give $o(1)$ in the ordered limit, not zero at finite cutoffs.
- The combined principal parts still leave $-19\zeta(4)Y/(2\pi)$.

Second curvature: the Regge condition

The massive-pole principal parts approach $-19\zeta(4)/2$ on the large imaginary axis. Regge decay fixes the compensating amplitude constant:

$$\int_{-iY}^{iY} \frac{dS}{2\pi i} \frac{19}{2} \zeta(4) = \frac{19\zeta(4)Y}{2\pi}.$$

This is an amplitude constant, not a missing simple-pole residue.

$$\begin{aligned} I^{(2)}(0) &= \frac{263}{6} \zeta(4) + 6\zeta(2) + \left(\zeta(2) - \frac{13}{2} \right) \zeta(3) - \frac{5}{3} \zeta(5) \\ &= 49.7471610332818994 \dots \end{aligned}$$

- This is the ordered limit with the Regge completion included.
- Even at $Y = 10^6$, the completed contour gives $49.7282427441 \dots$
- Numerical checks of subtracted residue remainders do not establish the convergence rate of the finite-axis integral.